American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

Project Title : Street Children Care System

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Submitted By

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Assessment Rubric

CO1: Explain the impact of software engineering over various contexts of software development to assess societal, health, safety, legal and cultural issues.

Student ID	Problem Analysis [3]	Socio-cultural Impact [3]	Related Solutions and Studies [4]
1. 23-51771-2			
2. 23-51788-2			
3. 23-51745-2			
4. 24-56230-1			
5. 21-45435-3			

CO2: Explain software engineering principles considering professional responsibilities and societal impacts.

Student ID	Conceptual Understanding [3]	Principle Selection and Justification [3]	Application in Software Development Context [4]
1. 23-51771-2			
2. 23-51788-2			
3. 23-51745-2			
4. 24-56230-1			
5. 21-45435-3			

Table of Contents

1. Project Proposal 01
   1.1 Background to the Problem 01
   1.2 Selection of Process Model 02
2. Timeline Chart / Gantt Chart……..04
3. Software Requirements Specification (SRS) / PRD 05
   3.1 Scopes and Features 05
   3.2 User Story Table 08
   3.3 Requirements Traceability Matrix 13
      3.3.1 Functional Requirements 13
      3.3.2 Non-Functional Requirements 17
4. Software Design 18
   4.1 Use Case Diagram 18
   4.2 UI Design using Figma……..……..20
5. Software Testing 29
6. Git Workflow 31
7. Conclusion 36

1. PROJECT PROPOSAL

1.1 Background to the Problem

There are many street children in Bangladesh who do not have a fixed place to stay, ID card or access to health facilities. These children are primarily kept on paper-based manual records, though organizations like orphanages, NGOs and charitable foundations attempt to register and provide aid to these children. With these paper systems, there is no real control over who can access sensitive child records, missed vaccination dates, duplicate entries, and lost records. For instance, a staff member could have to navigate through a couple of paper files only to determine whether a child has already had a vaccination, or two different offices may have separate and conflicting records for a child.

The key issue is not that these organisations are not caring, rather it is that there is no single system that links children, orphanages, foundations and health records together. If child records, orphanage information, information about the foundation's support for the child, and vaccination history are kept on paper and are separated, issues like duplicate data, missing data, and human error start to become common. It may take a long time to trace a child's record or to determine the orphanage or foundation to which he or she is attached, or to ascertain if he or she has been vaccinated. An effective solution needs to provide current child data, easy logon for each user, and instant access to health records and services at all times. The solution to these problems in this project is the development of a Street Children Care System with C# Windows Forms as front-end and SQL Server as back-end database.

A centralized system would be suitable for this project as it will allow for child registration, orphanage information, foundation support and vaccination tracking to be housed in one location. All the records are stored in one unified database, minimising the need for duplicate paper work, lost files and ensuring that everyone has up to date information at the same time. Login and role based access ensure that only those users with the appropriate role can view or edit only information pertaining to their role. The target of this project is to create a centralized system for the care of street children designed to register children, connect them to orphans and foundations and follow-up their vaccine history; all in one secured, user-friendly platform.

The Street Children Care System hopes to minimize data loss, lessen the risk of human error, secure sensitive data of street children and the organizations that provide care, and provide for more timely and accurate assistance for street children and the organizations that care for them, replacing the need for paper-based and scattered data.

Target Group of Users

The system will be used by five main groups of users:

(i) Administrator

The entire system is managed by the Administrator, who will have complete control of the platform. The Administrator establishes user accounts, assigns roles and is responsible for all children, orphanage, foundation and vaccination records, as well as the security and correctness of all systems.

(ii) Staff

The registration and case management of street children are taken by staff every day. Staff can add a new child's record and update the status or personal details of a child and be able to search or sort children by name, age and status to quickly find information.

(iii) Orphanage Coordinator

Orphanage Coordinators are the representatives in the Orphanages where street children are put up. They maintain their own information including name, location and contact number and track down the children who are placed under their care.

(iv) Foundation Coordinator

Foundation Coordinators are the person who represents the foundations supporting the street children with food, clothing, education, or medical assistance. They handle the details and contact information of their foundation, and maintain records of the children their foundation supports.

(v) Vaccination Staff

Vaccination Staff take care of all children's health records. They add new vaccinations, change the status of each vaccination (pending, completed, or overdue) and search vaccinations to identify children that require a pending or missed vaccination.

The Street Children Care System offers a single platform that is secure, centralized and user-friendly for these five groups of users to do their daily work. The system's objective is to provide minimal handwriting, child protection, and ensure that all street children are registered, cared for and monitored in their health.

1.2 Selection of Process Model

Our software process model for the Street Children Care System was selected to be Agile - Extreme Programming (XP). XP is an Agile based development model to facilitate iterative development, continuous feedback, team collaboration, high software quality by delivering small releases. The aim of our desktop application is to allow the management of five different roles: Administrator, Staff, Orphanage Coordinator, Foundation Coordinator and Vaccination Staff. The collaboration between the teams happened in weekly iterations, tasks were managed using Trello, code was

reviewed in the form of PR (pull request) in Git and there was the flexibility to make any future changes to requirements; all these were a perfect match with our real workflow.

Process Model Phases:

(i) Exploration: We found our initial complaints about manual child record keeping, determined our functional boundaries for the 5 user roles and chose our tech stack.

(ii) Planning: User stories were written for every 5 roles, iteration goals were established, and a Trello board was set up to keep track of tasks in their To Do, In Progress, Review and Done statuses.

(iii) Iteration to Release: Development was broken up into short weekly iterations. Team members developed their own branches on Git and wrote module code before submitting a Pull Request (PR) for team review, then merged into the main branch.

(iv) Productionizing: Integrated system testing of all modules to ensure data connections, role-based access security and accuracy of data before submission.

(v) Maintenance: This is where UI tweaks and bug fixes are done and the database is continuously updated if future user needs arise.

(vi) Death: When the project ends up at full documentation submission and software hand-off, no new feature iterations added, or at the end of the work product presentation.

Here is a simple diagram to show the **XP process:**

Exploration
↓
Planning
↓
Iterations to Release
↓
Productionizing
↓
Maintenance
↓
Death

Why Extreme Programming (XP) is the Best Fit:

(i) Weekly Iterative Delivery: This involved dividing up the project into tasks, and doing them in small chunks each week, directly following the XP principle of iteration.

(ii) XP values collective code ownership and continuous integration: Trello & Git Code Reviews. High code quality across all 5 team members was maintained with task statuses on Trello and code review using Git Pull Requests.

(iii) Flexibility for requirement changes: XP helps to easily make changes to features or changes to the user interface in subsequent changes if operations need to evolve.

(iv) Effective Team Collaboration: Working together on shared Google Workspace documents and maintaining regular communication kept all user modules synchronized.

(v) 40-hour week: Developers can maximum 40-hour work in a week. So that the developers' working productivity will increase.

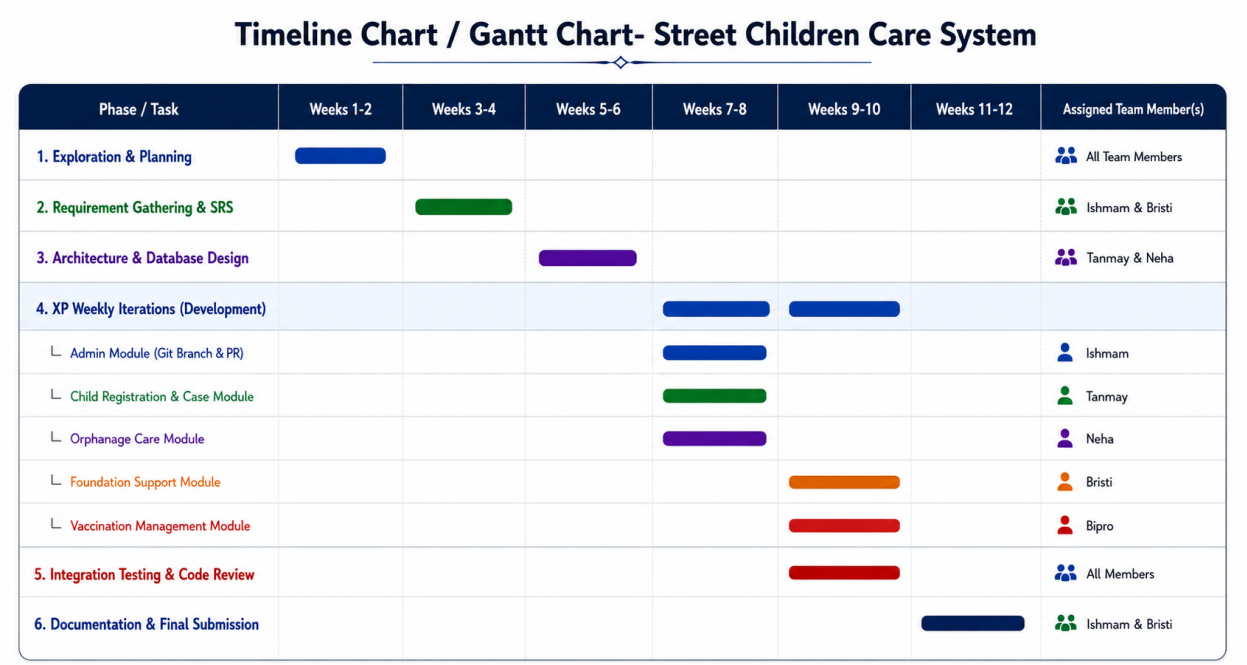
Extreme Programming (XP) will work perfectly in our Street Children Care System because core XP values like communication, simplicity, courage, and respect will enhance our team collaboration across all five user roles. Besides, its practices-such as weekly iterations, Trello task tracking, and Git code reviews will help us develop the system smoothly.

2. Timeline Chart / Gantt Chart

Team Member Task Allocation

- **Ishmam (Administrator):** Managed project workflow, Trello board and GitHub repository; prepared project documentation, designed the master Figma layout and built the core Admin modules.
- **Tanmay (Staff):** Conducted requirement analysis, collaborated on GitHub and Trello tasks and built the Street Child Registration and Case Management Module.
- **Neha (Orphanage Coordinator):** Designed database architecture, collaborated on GitHub and Trello tasks and built the Partner Orphanage Care and Child Tracking Module.
- **Bristi (Foundation Coordinator):** Assisted with documentation work, contributed to GitHub and Trello tasks and built the Foundation Support and Donation Management Module.
- **Bipro (Vaccination Staff):** Vaccination Record Management Module, Health Drive Scheduling, System Testing.

Timeline Chart / Gantt Chart- Street Children Care System



3. SOFTWARE REQUIREMENTS SPECIFICATIONS (SRS)

3.1 Scopes and Features

Street Children Care System is a management system that is centralized in a desktop computer to manage street children operations based on five main users.

Scope 1: User Authentication and Access Management

Features:

a) Limits system modules, options and data access by five user roles pre-defined (Administrator, Staff, Orphanage Coordinator, Foundation Coordinator, Vaccination Staff).

b) Authenticates new user registrations using mobile OTP and user confirmation email.

c) Securely logs on to the system with the username and password of the registered user.

Scope 2: Account Enrollment and Provisioning

Features:

a) Allows Administrators to create new profiles for Staff, Orphanage Coordinators, Foundation Coordinators and Vaccination Staff.

b) Enables Administrators to change user access, edit user account information or remove access to the system for users.

c) Lists all the users of the system enrolled and their account details in a searchable list for Administrators.

Scope 3: Self-Profile and Credential Management

Features:

a) Provides access to individual user's details, as they have entered into the system database, to all users who are logged in.

b) Allows users to edit personal contact information or personal profile whenever it's needed.

c) Offers a safe way for users to reset their passwords by requiring them to enter their old password and then their new one, and also sending an OTP to the mobile phone and email to confirm the new password when it is forgotten.

Scope 4: Street Child Registration

Features:

a) To allow Staff to insert new street children into the database with demographic data, estimated age, gender, location and initial condition of physical condition.

b) Validates required input fields before adding a new child record to ensure that no child record gets added that's incomplete.

Scope 5: Child Case Management and Tracking

Features:

a) Provides a master list of registered street children to approved Staff and Administrators.

b) Enables search and filtering of child records by name, age and current care status.

c) Allows Staff and Administrators to make changes to any registered child's personal information, health condition and placement progress over time.

Scope 6: Orphanage Partner Management

Features:

a) Enables Administrators to insert new partner orphanages to centralized system databases.

b) Enables Administrators to view, edit and manage all the details of the registered Orphanage Partners.

Scope 7: Orphanage Care Coordination

Features:

a) Allows Orphanage Coordinators to keep all the information regarding their Orphanage, such as its profile, location, capacity and representative contact numbers.

b) Keeps and presents information on every street child being housed in particular orphanages.

Scope 8: Foundation Partner Management

Features:

a) Enables Administrators to sign up new street children partner charitable foundations.

b) Allows Administrators to add, edit and review the list of registered charitable foundations.

Scope 9: Foundation Support Coordination

Features:

a) Enables Foundation Coordinators to manage organizational details, contact information, and types of support provided, such as food, clothing, education, or healthcare.

b) Maintains records that link supported street children directly to their respective donor foundations.

Scope 10: Health and Immunization Record

Features:

a) Provides Vaccination Staff with a means to document the vaccines given to each child, the dosage and the date of the vaccine.

b) Automatically computes and updates child vaccination statuses (pending, completed, and overdue).

c) Enables the Vaccination Staff to make edits and deletions to incorrect health records to provide a proper medical history.

Scope 11: Health Drive Scheduling and Monitoring

Features:

a) Enables Vaccination Staff to create and search lists of children who are pending or overdue for vaccinations.

b) Generates immunization tracking data which will guide efficient planning and scheduling of field vaccination visits.

3.2 User Story Table

As A/An	I want to	So that	Acceptance Criteria
Administrator	Create a new Administrator profile	I can enroll myself in the system	An OTP is sent to user's mobile and confirmation mail to the registered email
	Create a new staff profile	I can enroll a staff in the system	An OTP is sent to staff's mobile and confirmation mail to the registered email
	Create a new Orphanage Coordinator profile	I can enroll an orphanage coordinator in the system	An OTP is sent to orphanage coordinator's mobile and confirmation mail to the registered email
	Create a new Foundation Coordinator profile	I can enroll a foundation coordinator in the system	An OTP is sent to foundation coordinator's mobile and confirmation mail to the registered email
	Create a new Vaccination Staff profile	I can enroll vaccination staff in the system	An OTP is sent to vaccination staff's mobile and confirmation mail to the registered email
	Login with my username and password	The system can authenticate me and I can trust it	After successful login, the data associated to the user should be accessible
	View my own profile details	I can see a list that contains information provided in the registration form	Information list table window including edit profile button
	Update my information	I can replace the incorrect information with the right information	Information list table window including edit profile button
	Change my password	My profile can be more secure	User must input old password, new password twice and given notification of success

	Request a new password	I do not permanently lose access to my data if I forget the password	An OTP is sent to user's mobile and confirmation mail to the registered email
	Remove user profile	The user cannot access the system	Will show a popup window with details and ask for confirmation
	View system users list	I can check who is using the system	User list table with status and roles
	Add new orphanage	I can register a new orphanage partner	User must input required information and given notification of success
	View orphanage list	I can see all registered orphanages	Show updated orphanage list with update and delete button
	Add new foundation	I can register a new foundation partner	User must input required information and given notification of success
	View foundation list	I can see all registered foundations	Show updated foundation list with update and delete button
	Assign child to orphanage	The child is placed under proper care	Show updated child list linked to the orphanage
	Assign child to foundation	The child receives support	Show updated child list linked to the foundation
	View vaccination status report	I can check child vaccination progress	Show report table of completed, pending, and overdue vaccinations

Staff	Login with my username and password	The system can authenticate me and I can trust it	After successful login, the data associated to the user should be accessible
	View my own profile details	I can see a list that Contains information provided in the registration form	Information list table window including edit profile button
	Update my information	I can replace the incorrect information with the right information	Information list table window including edit profile button
	Change my password	My profile can be more secure	User must input old password, new password twice and given notification of success
	Request a new password	I do not permanently lose access to my data if I forget the password	An OTP is sent to user's mobile and confirmation mail to the registered email
	Add new child profile	I can register a new street child	User must input required information and given notification of success
	Search for child	I can find a specific child quickly	Show list of desired children records
	View child list	I can see the details and status of all children	Show updated child list with update and delete button
	Update child status	I can change placement status	Show updated child list with new status
Orphanage Coordinator	Login with my username and password	The system can authenticate me and I can trust it	After successful login, the data associated to the user should be accessible

	View my own profile details	I can see a list that contains information provided in the registration form	Information list table window including edit profile button
	Update my orphanage information	The orphanage details stay correct	Information list table window including edit profile button
	Changed my password	My profile can be more secure	User must input old password, new password twice and given notification of success
	Request a new password	I do not permanently lose access to my data if I forget the password	An OTP is sent to user's mobile and confirmation mail to the registered email
	Search for child	I can find a specific child quickly	Show list of desired children records
	View children under my care	I can see all children in my orphanage	Show children list table with total count of children
Foundation Coordinator	Log in using my user name & password	I can control my foundation's information in the system	After successful login, the user is redirected to the dashboard & can access all information
	Update my foundation's details and contact information	The foundation's information always remains accurate and up-to-date.	Information is displayed in a table & an edit profile option is available
	View my own profile details	I can see a list that contains information provided in the registration form	Information list table window including edit profile button

	View the list of children my foundation supports	I can keep track of every children	A list of children's information linked to my foundation is shown
	Search for a specific child my foundation supports	I can find that child's record quickly	The child's record is displayed on screen when there is a match.
	Change my password	My profile can be more secure	User must input old password, new password twice and given notification of success
	Request a new password	I do not permanently lose access to my data if I forget the password	An OTP is sent to user's mobile and confirmation mail to the registered email
	Record & manage donation	I can keep track of all donations accurately	Donation details are saved successfully & can be viewed & updated
	Update funding information	Financial records remain accurate	Changes are saved in the system
Vaccination Staff	Login with my username and password	The system can authenticate me and I can trust it	After successful login, the data associated to the user should be accessible
	View my own profile details	I can see a list that contains information provided in the registration form	Information list table window including edit profile button
	Update my information	I can replace the incorrect information with the right information	Information list table window including edit profile button
	Change my password	My profile can be more secure	User must input old password, new password twice and given notification of success

	Request a new password	I do not permanently lose access to my data if I forget the password	An OTP is sent to user's mobile and confirmation mail to the registered email
	Add child vaccination record	The child vaccination history stays correct	User must input required information and given notification of success
	Update vaccination status	The system shows updated status	Show updated vaccination list with new status
	Remove vaccination record	Incorrect record does not remain	Show updated list without that record
	View pending or overdue vaccinations	I can plan the next vaccination visit	Show list of children with pending or overdue vaccinations

3.3 Requirements Traceability Matrix

3.3.1 Functional Requirements

1.0	**System Administrator Requirements**
1.1	**Registration and User Enrollment**
1.1.1	Taking user role selection (Admin, Staff, Orphanage Coordinator, Foundation Coordinator, Vaccination Staff)
1.1.2	Taking user profile details
1.1.3	Sending an OTP to the user's mobile device
1.1.4	Sending a confirmation email to the registered email address
1.1.5	Submitting the enrollment form and showing a notification of success
1.2	**Login**
1.2.1	Taking username
1.2.2	Taking password

1.2.3	Providing a request new password option
1.2.4	Submission of the login credentials
1.3	**Dashboard**
1.3.1	Side menu
1.3.1.1	Dashboard button
1.3.1.2	View profile button
1.3.1.2.1	Show registration information list
1.3.1.2.2	Edit profile button
1.3.1.3	Change password button
1.3.1.3.1	Taking old password
1.3.1.3.2	Taking new password twice
1.3.1.3.3	Showing notification of success
1.3.1.4	User Management button
1.3.1.4.1	View system users list table with status and roles
1.3.1.4.2	Remove user profile
1.3.1.4.3	Show confirmation popup window for removal
1.3.1.5	Organization Management button
1.3.1.5.1	Add new orphanage
1.3.1.5.2	View updated orphanage list with update and delete buttons
1.3.1.5.3	Add new foundation
1.3.1.5.4	View updated foundation list with update and delete buttons
1.3.1.6	Child Placement button
1.3.1.6.1	Assign child to orphanage
1.3.1.6.2	Assign child to foundation
1.3.1.7	Reports button

1.3.1.7.1	View vaccination status report (completed, pending, overdue)

2.0	**Staff Requirements**
2.1	**Login**
2.2	**Dashboard**
2.2.1	Side menu
2.2.1.1	View profile button
2.2.1.2	Update information button
2.2.1.3	Change password button
2.2.1.4	Child Case Management button
2.2.1.4.1	Add new child profile
2.2.1.4.2	Search for child
2.2.1.4.3	View child list with update and delete buttons
2.2.1.4.4	Update child placement status

3.0	**Orphanage Coordinator Requirements**
3.1	**Login**
3.2	**Dashboard**
3.2.1	Side menu
3.2.1.1	View personal profile details
3.2.1.2	Update orphanage information
3.2.1.3	Change password button
3.2.1.4	Child Tracking button
3.2.1.4.1	Search for child

3.2.1.4.2	View children under my care

4.0	**Foundation Coordinator Requirements**
4.1	**Login**
4.2	**Dashboard**
4.2.1	Side menu
4.2.1.1	View personal profile details
4.2.1.2	Update foundation's details and contact information
4.2.1.3	Change password button
4.2.1.4	Supported Children button
4.2.1.4.1	View the list of children my foundation supports
4.2.1.4.2	Search for a specific child my foundation supports
4.2.1.5	Finance Management button
4.2.1.5.1	Record and manage donation
4.2.1.5.2	Update funding information

5.0	**Vaccination Staff Requirements**
5.1	**Login**
5.2	**Dashboard**
5.2.1	Side menu
5.2.1.1	View profile details
5.2.1.2	Update information
5.2.1.3	Change password button
5.2.1.4	Vaccination Management button

5.2.1.4.1	Add child vaccination record
5.2.1.4.2	Update vaccination status
5.2.1.4.3	Remove incorrect vaccination record
5.2.1.4.4	View list of children with pending or overdue vaccinations

3.3.2 Non-Functional Requirements

1.0	**Security & Access Control**
1.1	The system must ensure that only authenticated users can access child data and organizational records based on their assigned role.
1.2	Passwords must be updated using a secure two-step verification process requiring the old password and double entry of the new password.
1.3	Password recovery must be secured using a dual-channel verification method (Mobile OTP and Email Confirmation).
2.0	**Data Integrity & Reliability**
2.1	The system must operate on a centralized SQL Server backend database to instantly prevent duplicate child entries and lost records.
2.2	Removing records (such as deleting a user profile or a vaccination record) must prompt a confirmation window to prevent accidental data loss.
3.0	**Usability & User Experience**
3.1	The system front-end must be built with C# Windows Forms to provide a familiar and simple desktop interface for staff and coordinators .
3.2	All data tables (such as child lists, vaccination reports, and user lists) must visually display status updates instantly upon modification.
3.3	The system must display clear success notifications and pop-ups after any major data entry (adding a child, updating a password, recording a donation).
4.0	**4.0 Performance & Efficiency**
4.1	The search function for child records must return results quickly to eliminate the delays previously caused by manual paper-file navigation.
4.2	System dashboards must load immediately after login, automatically filtering

	information so that users only see data relevant to their specific orphanage, foundation, or role.

4. SOFTWARE DESIGN

4.1 Use Case Diagram:

The Use Case Diagram represents a high-level view of the functional requirements and interactions of actors in the Street Children Care System. It visualizes the interactions between five core actors (Administrator, Staff, Orphanage Coordinator, Foundation Coordinator, Vaccination Staff) in the system and how they perform the basic activities (child registration, institutional allocations, tracking financial donations, and healthcare management).

(i) Administrator: User management of system users, role permission, register partner orphanages and foundations, etc. and review system reports overall.

(ii) Staff: Registers street children and keeps up to date with ongoing records of children's personal, demographic information.

(iii) Orphanage Coordinator: Maintains profile information of the Orphanage, keeps track of the overall capacity of the houses and monitors the street children being served by the Orphanage.

(iv) Foundation Coordinator: Maintains the information about the foundation and records financial or material donations and monitors children receiving educational or basic support.

(v) Vaccination Staff: Records doses of vaccine given, updates vaccine status of children (pending, completed, overdue) and arranges future health drives.

Figure 1: Use Case Diagram – Role-Based Actor Generalization and System Boundaries for Street Children Care System

4.2 UI / Wireframe Design using Figma

A prototype of the Figma UI/UX design of the Street Children Care System. The interface is modern, clean, dark themed and focuses on usability and efficient data navigation with a persistent sidebar navigation and structured layout. The screens address critical operational areas such as role-specific dashboards, street child registration, foundation and orphanage profile management, donation monitoring and vaccination monitoring. Each view has interactive forms, clear data tables and intuitive call-to-action elements that ensure uniformity and smoothness in user experience regardless of the main user role.

Figma Link: Street_Children_Care_System – Figma

UI Design:

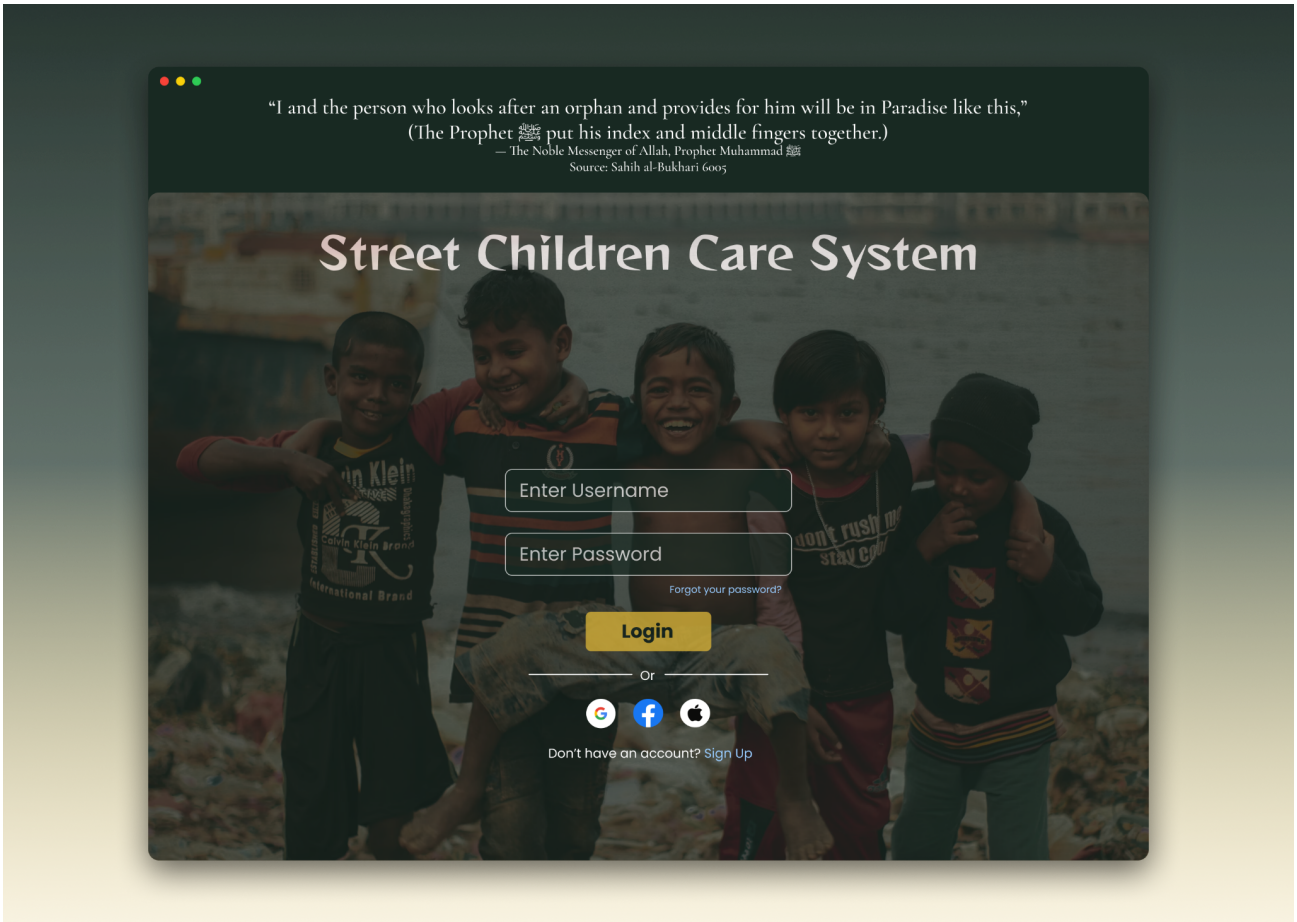


Figure No 1: Login (Ishmam)

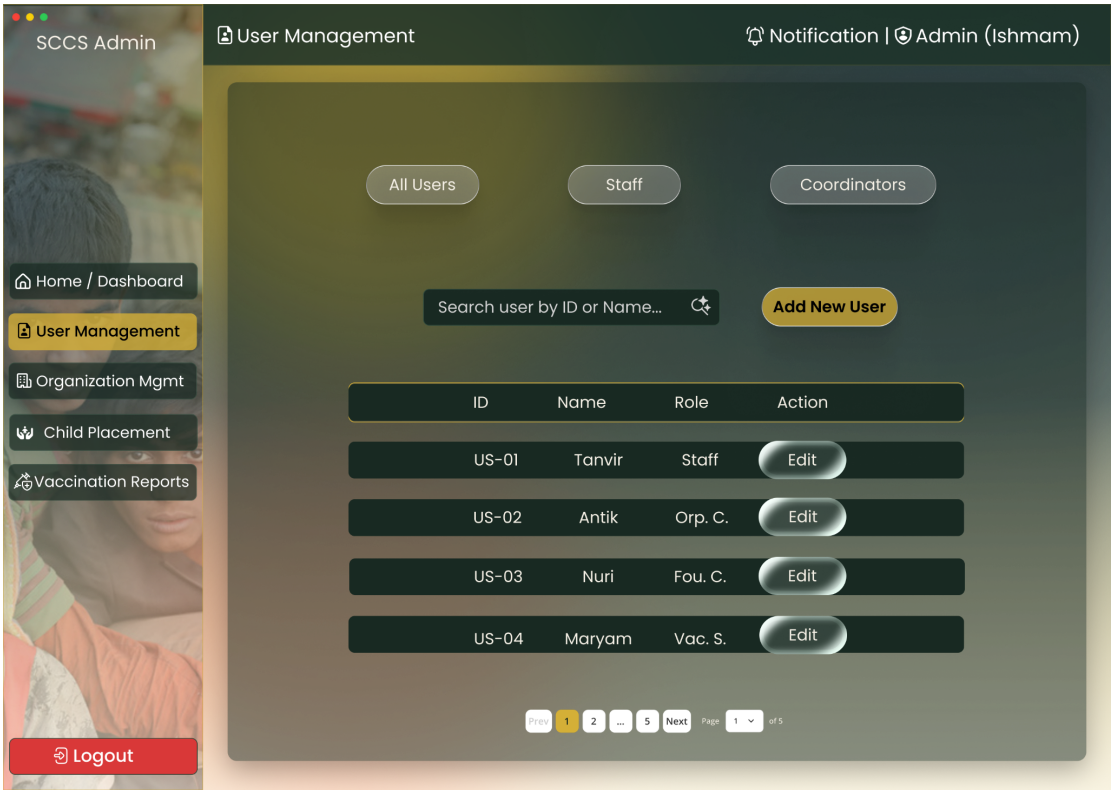


Figure No 2: User Management (Ishmam)

Figure No 3: Organization Management (Ishmam)

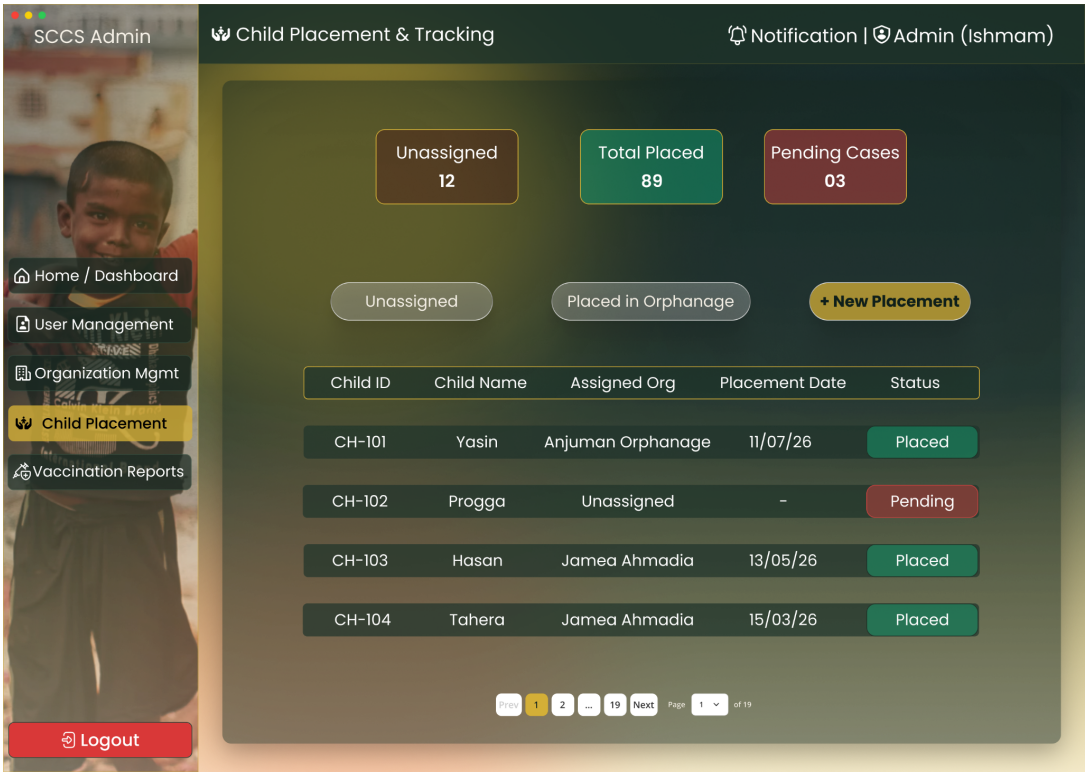


Figure No 4: Child Placement and Tracking (Ishmam)

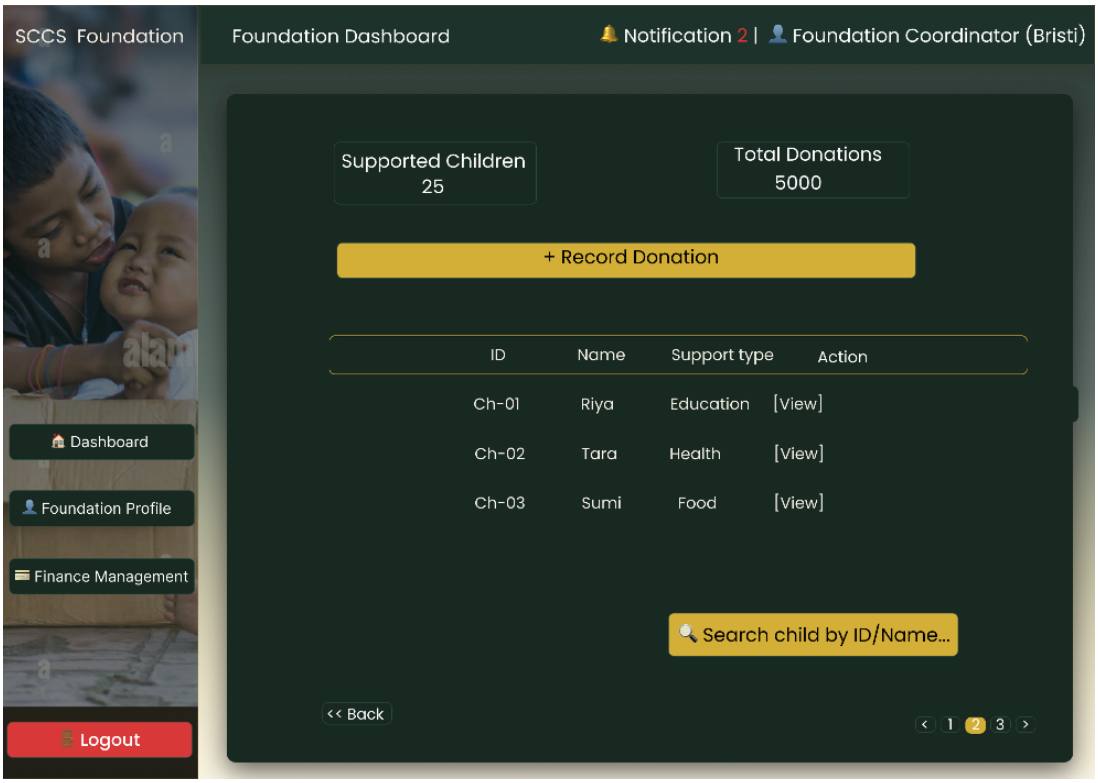


Figure No 5: Foundation Dashboard (Bristi)

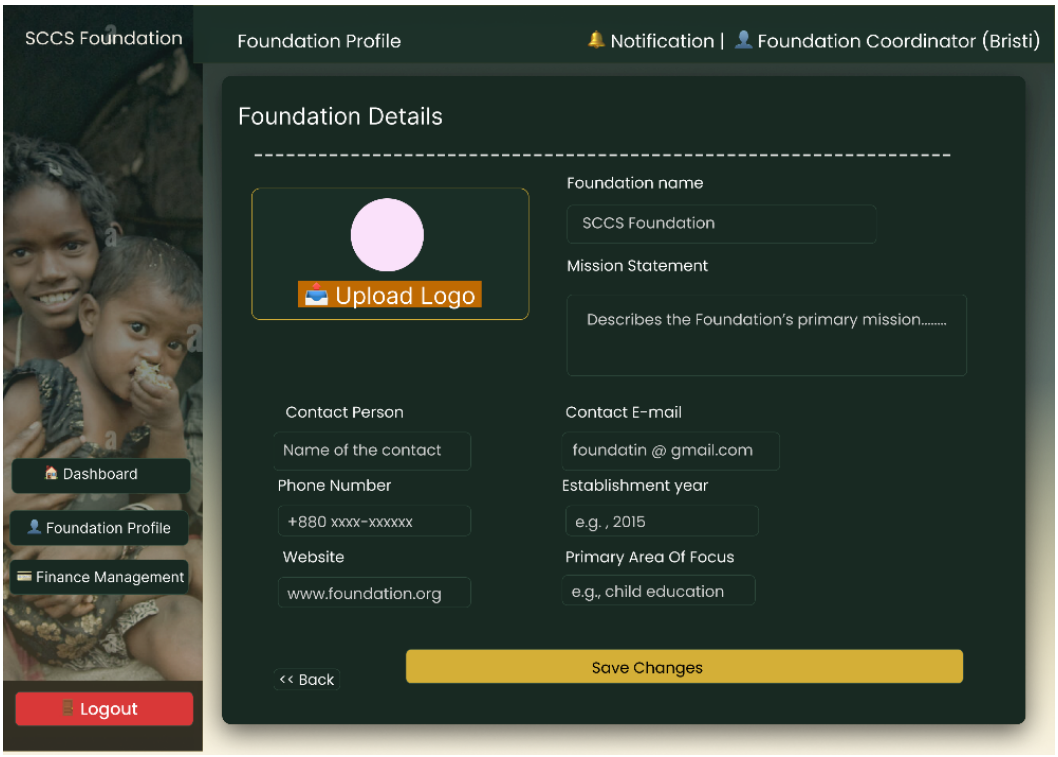


Figure No 6: Foundation Profile (Bristi)

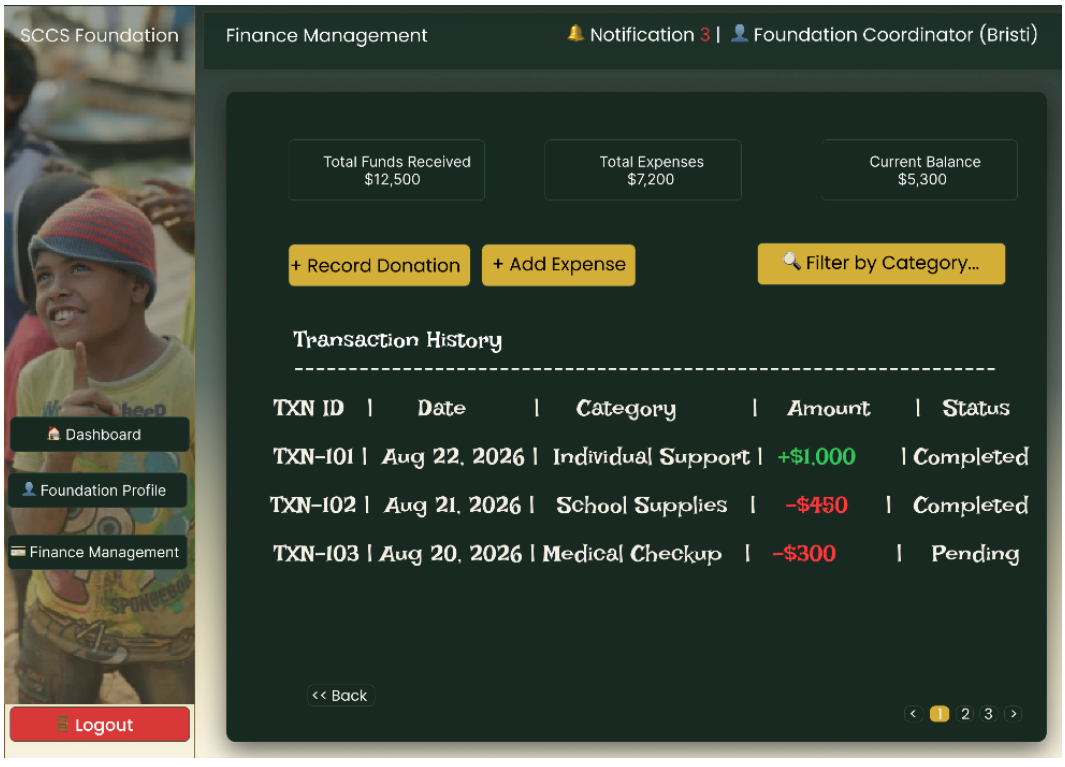


Figure No 7: Finance Management (Bristi)

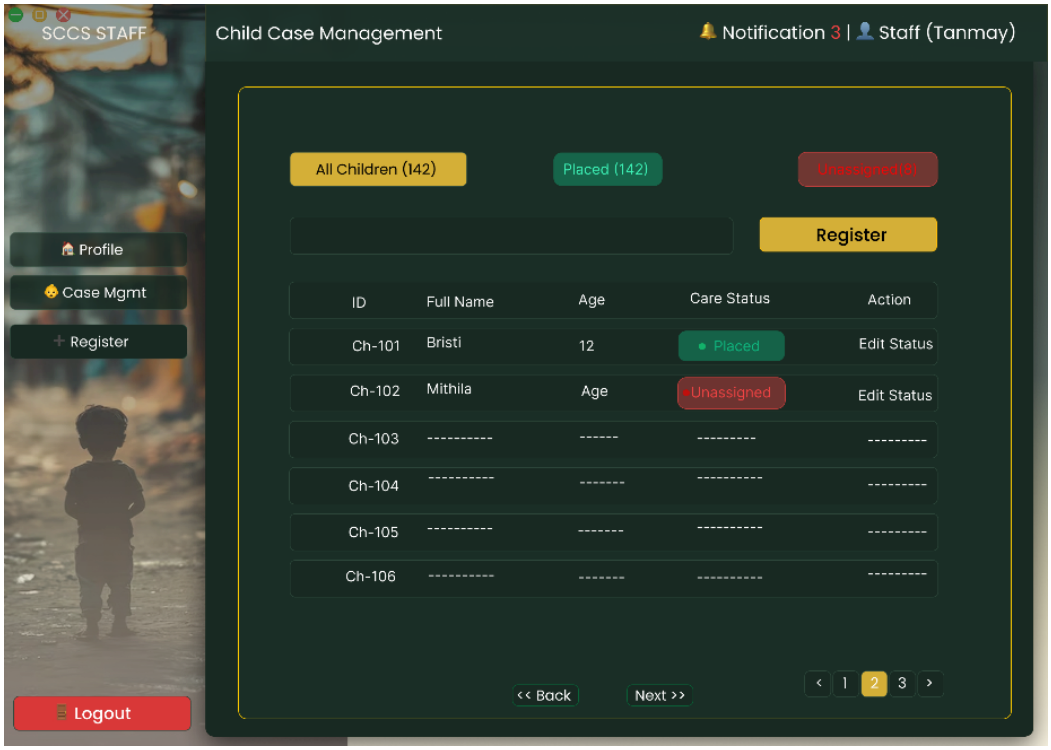


Figure No 08: Child Case Management (Tanmay)

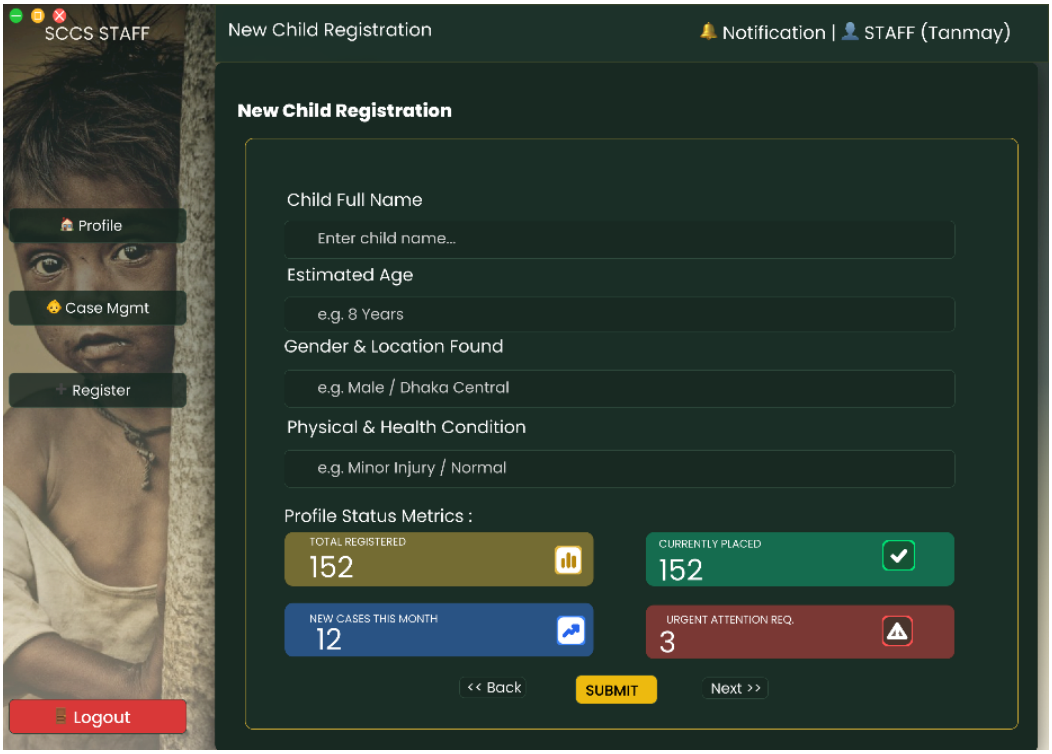


Figure No 09: New Child Registration (Tanmay)

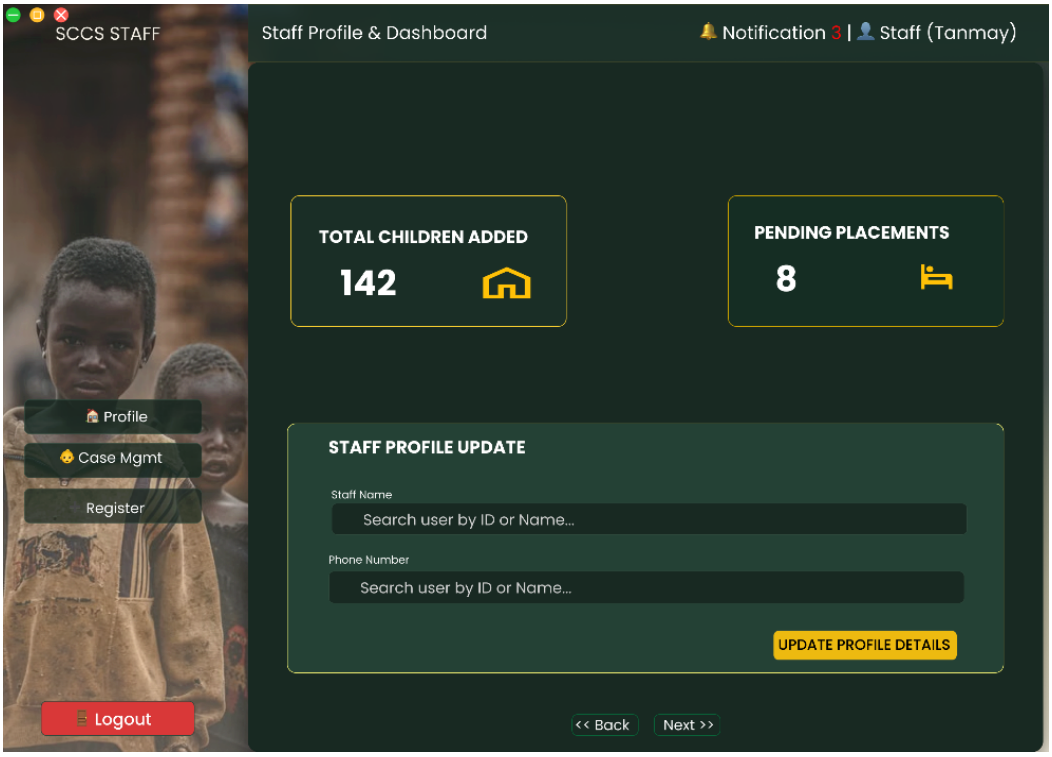


Figure No 10: Staff Profile & Dashboard (Tanmay)

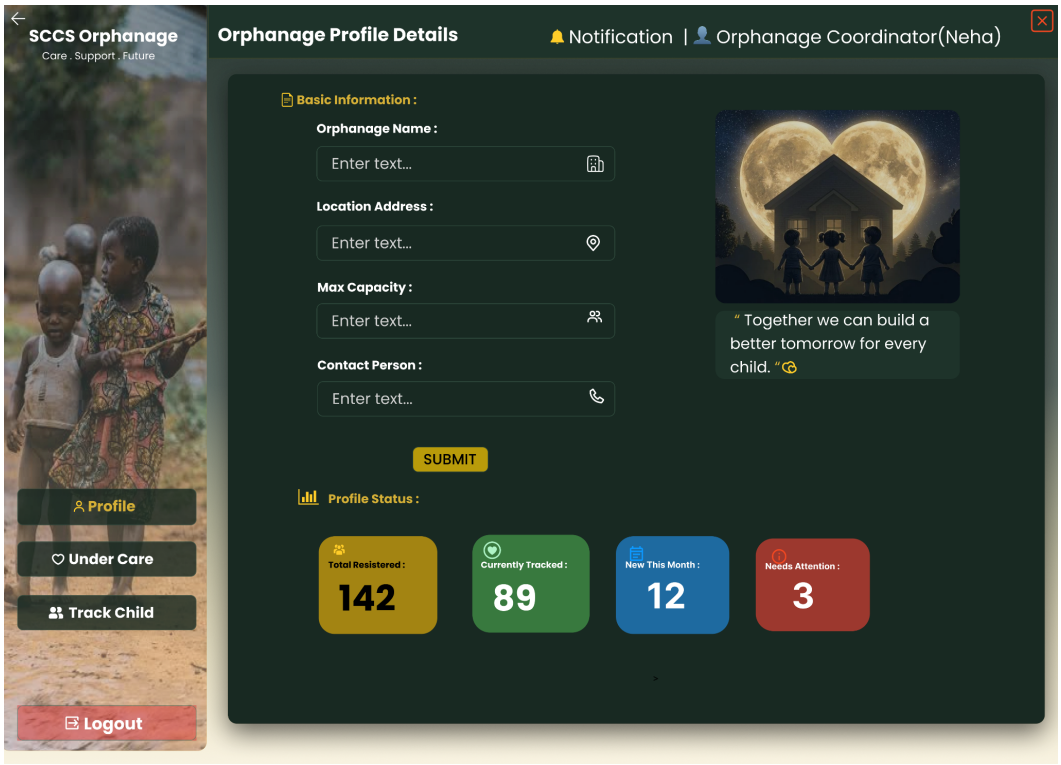


Figure No 11: Orphanage Profile Details (Neha)

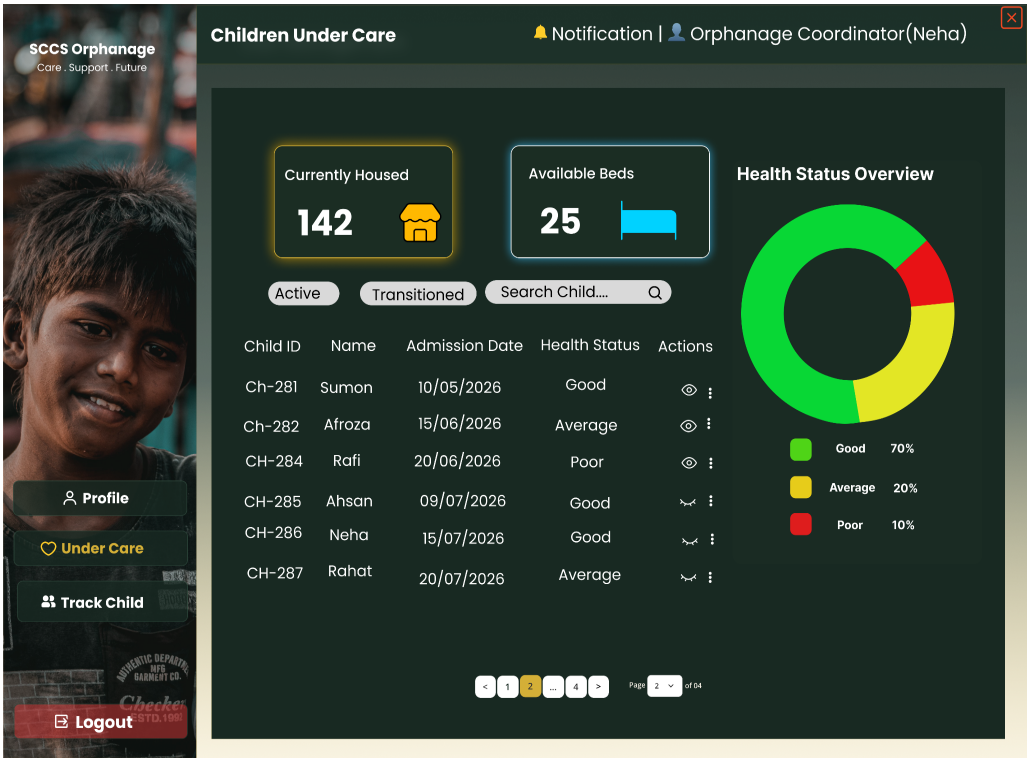


Figure No 12: Children Under Care (Neha)

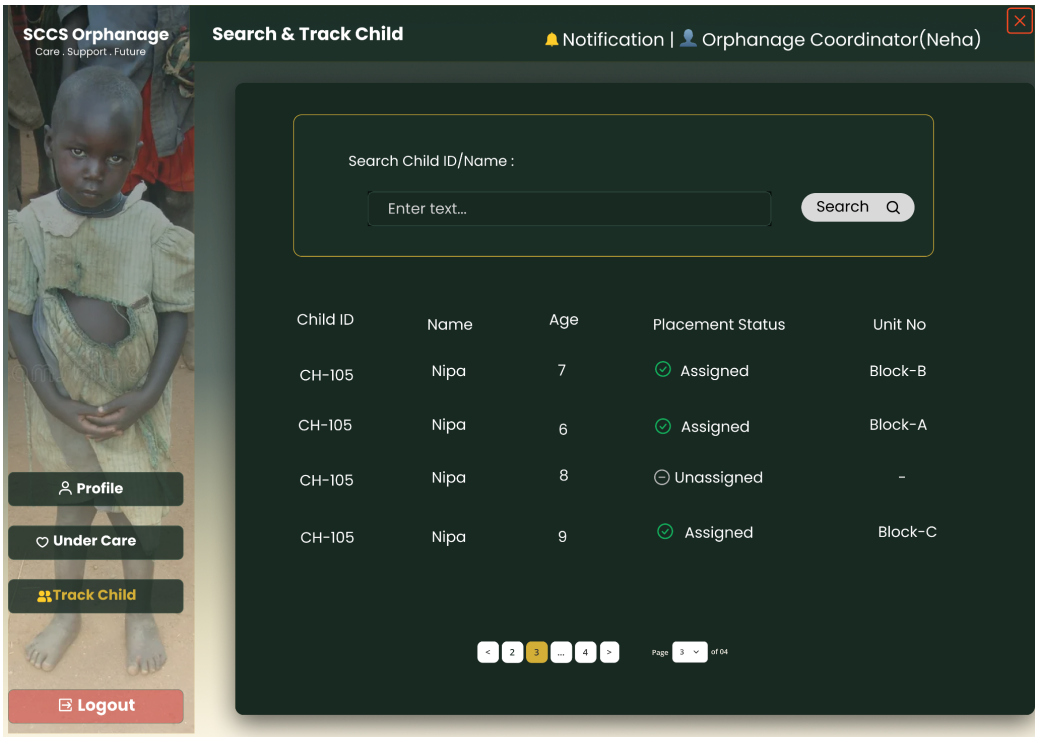


Figure No 13: Search & Track Child (Neha)

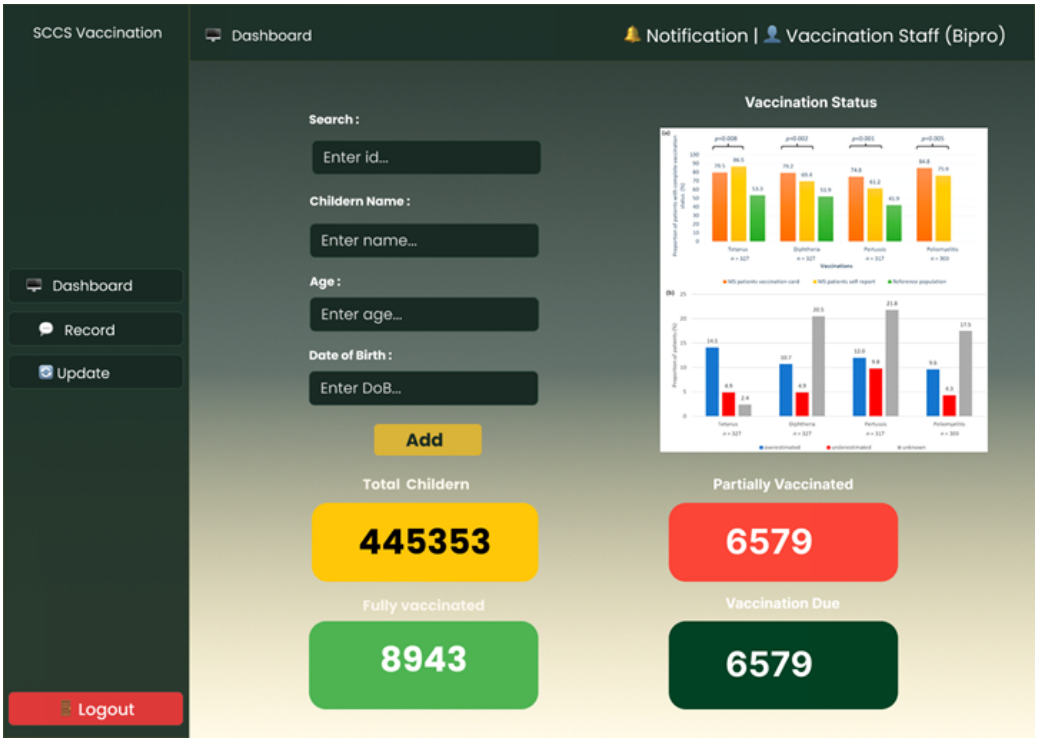


Figure No 14: Vaccination Dashboard (Bipro)

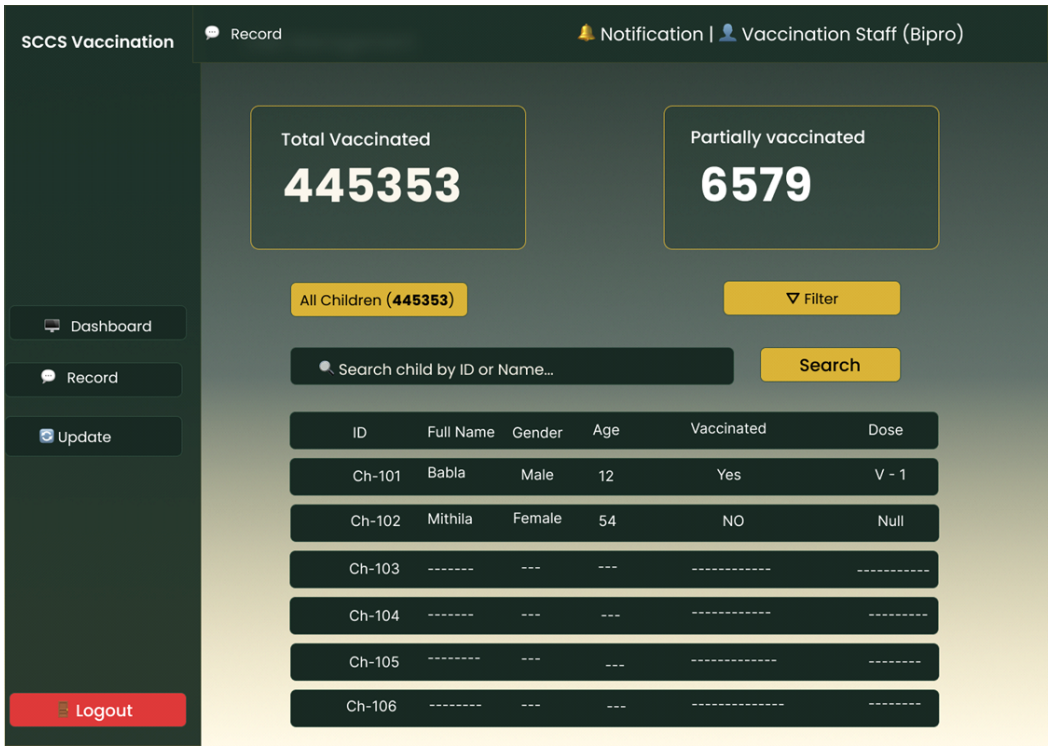


Figure No 15: Vaccination Record (Bipro)

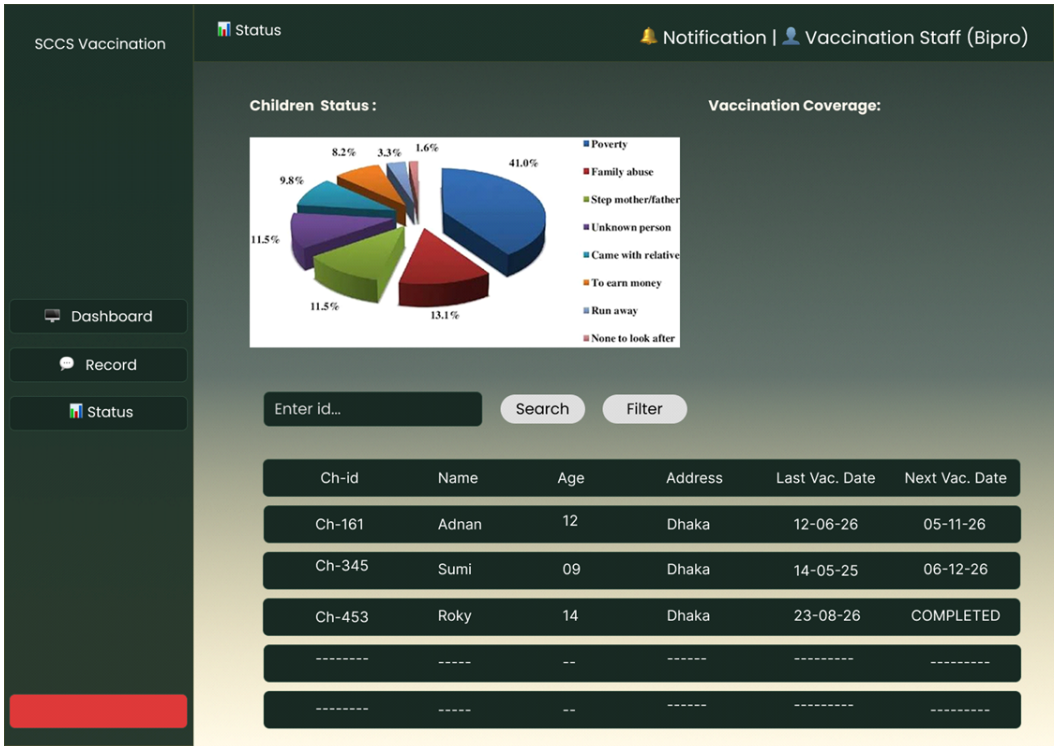


Figure No 16: Vaccination Status (Bipro)

5. SOFTWARE TESTING

Project Name: Street Children Care System			**Test Designed by: Ishmam Mubtasim**	
Test Case ID: TC_1			Test Designed date: 20/08/2026	
Test Priority (Low, Medium, High): High			Test Executed by: Ishmam Mubtasim	
Module Name: Child Placement & Tracking			Test Execution date: 22/08/2026	
Test Title: Verify Administrator assignment of an unassigned child to a registered orphanage				
Description: Test child placement module when Administrator assigns a child profile to a partner orphanage				
Precondition: Administrator is logged in and unassigned child profile exists in system				
Dependencies: Network connection and active database service				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Go to Child Placement module from side menu 2. Click "+ New Placement" button 3. Select Child ID from unassigned list 4. Select Target Organization from drop-down 5. Click submit	Child ID: CH-102 Child Name: Progga Assigned Org: Anjuman Orphanage Placement Date: 22/08/2026	Child status should update from "Pending" to "Placed" and reflect under Assigned Organization	As expected	Pass

Project Name: Street Children Care System	**Test Designed by: Sanjida Affrin Bristi**
Test Case ID: TC_2	Test Designed date: 27/08/2026
Test Priority (Low, Medium, High): High	Test Executed by: Sanjida Affrin Bristi
Module Name: Foundation Profile	Test Execution date: 27/08/2026
Test Title: Verify successful updating of foundation profile details with valid information	

Description: Test whether the Foundation Coordinator can successfully update and save foundation details. **Precondition: User is logged in as Foundation Coordinator and navigated to the Foundation Profile page.**				
Dependencies: Database connection to SQL Server must be active.				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Go to Foundation Profile page 2. Enter Foundation Name 3. Enter Contact Email & Person Name 4. Enter Phone Number & Website URL 5. Click 'Save Changes' button	Foundation Name: SCCS Foundation Email: foundation@contact.org Contact Person: Bristi Phone: +880 1700-000000	System validates input, updates record in SQL Server database and shows success notification.	Profile details updated successfully and notification displayed.	Pass

Project Name: Street Children Care System			**Test Designed by: Tanmay Roy Rony**	
Test Case ID: TC_3			Test Designed date: 27/08/2026	
Test Priority (Low, Medium, High): High			Test Executed by: Tanmay Roy Rony	
Module Name: New Child Registration			Test Execution date: 27/08/2026	
Test Title: Verify successful registration of a new street child with complete valid details				
Description: Test whether the Staff user can successfully register a new street child by entering all required information into the system. **Precondition: Staff user is logged into the system and navigated to the New Child Registration page.** **Dependencies: Database connection to SQL Server must be active.**				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Go to New Child Registration page	Name: Rahat Hossain Age: 8	System validates input, saves record to SQL	Record saved and success message displayed as expected.	Pass

2. Enter Child Full Name 3. Enter Estimated Age 4. Enter Gender & Found Location 5. Enter Physical & Health Condition 6. Click SUBMIT button	Gender/Location: Male, Karwan Bazar Health: Malnourished, minor fever	Server database, and shows a success notification.		

Project Name: Street Children Care System			**Test Designed by: Afroza Akter Neha**	
Test Case ID: TC_4			Test Designed date: 27/08/2026	
Test Priority (Low, Medium, High): High			Test Executed by: Afroza Akter Neha	
Module Name: Orphanage Profile Details			Test Execution date: 27/08/2026	
Test Title: Verify profile update failure when submitting non-numeric data in Max Capacity field				
Description: Test system validation behavior when an Orphanage Coordinator inputs invalid text into the numeric Max Capacity field. **Precondition: Orphanage Coordinator is on the Profile Details page.** **Dependencies: Input validation logic implemented on front-end C# Windows Forms.**				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Go to Profile page 2. Enter valid Orphanage Name 3. Enter invalid text in Max Capacity field 4. Click SUBMIT button	Name: Hope Home Shelter Max Capacity: Fifty Beds *(Invalid Text)*	System should block submission and display error notification: 'Max Capacity must be a positive integer.'	Form was submitted without validation and created a corrupted entry in database.	Fail

6. GIT WORKFLOW

Workflow Process:

1. **Central Repository** - Created on GitHub with main branch

2. **Clone Repository** - Each member clones the main repository
3. **Feature Branching** - Create individual branches for specific tasks
4. **Regular Commits** - Stage and commit changes with descriptive messages
5. **Push Updates** - Push changes to remote feature branches
6. **Pull Requests** - Create PRs for code review
7. **Merge to Main** - After approval, merge feature branches to main
8. **Continuous Integration** - Regular updates and conflict resolution

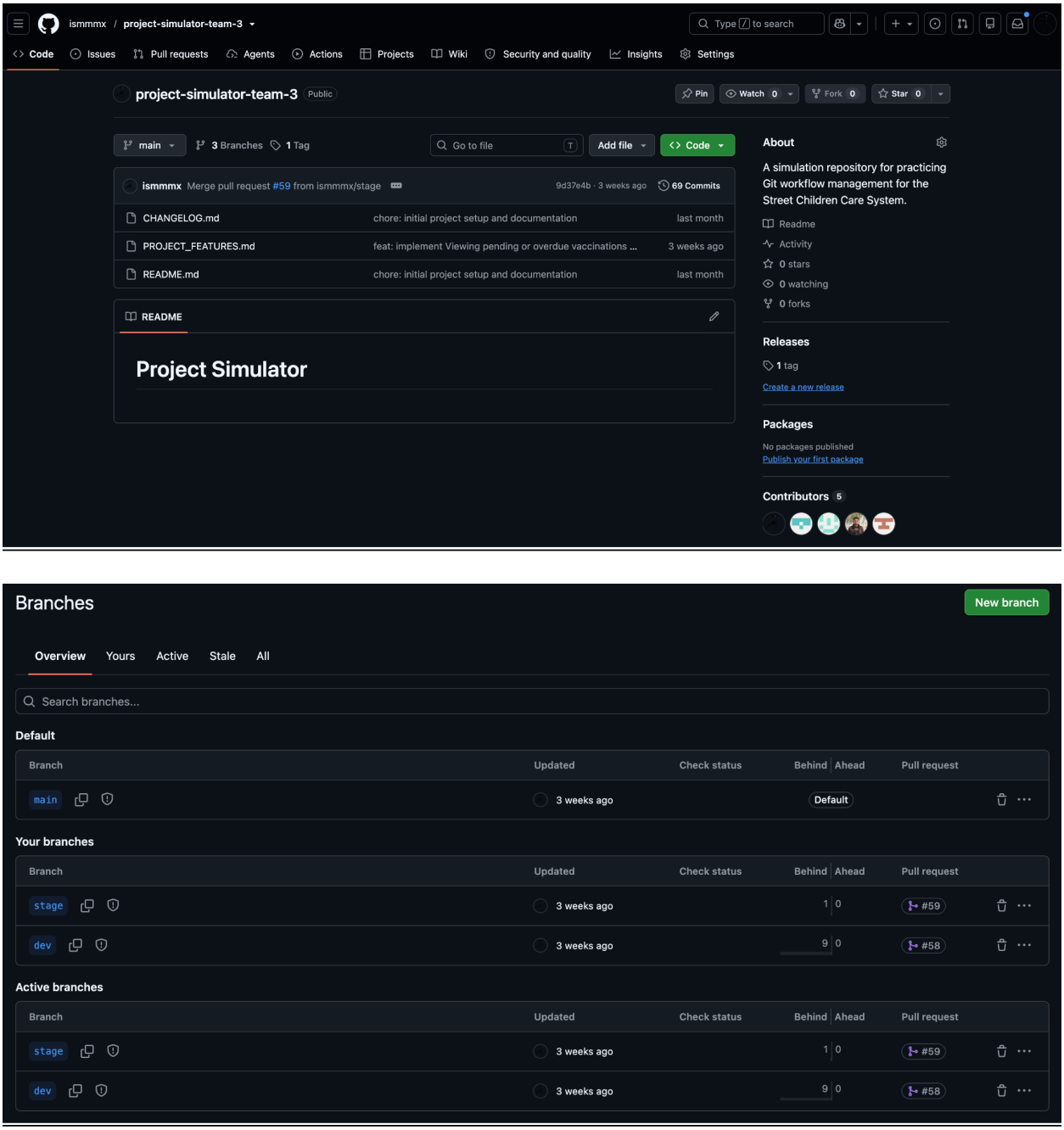


Figure 5.1: GitHub Central Repository with Active Feature Branches

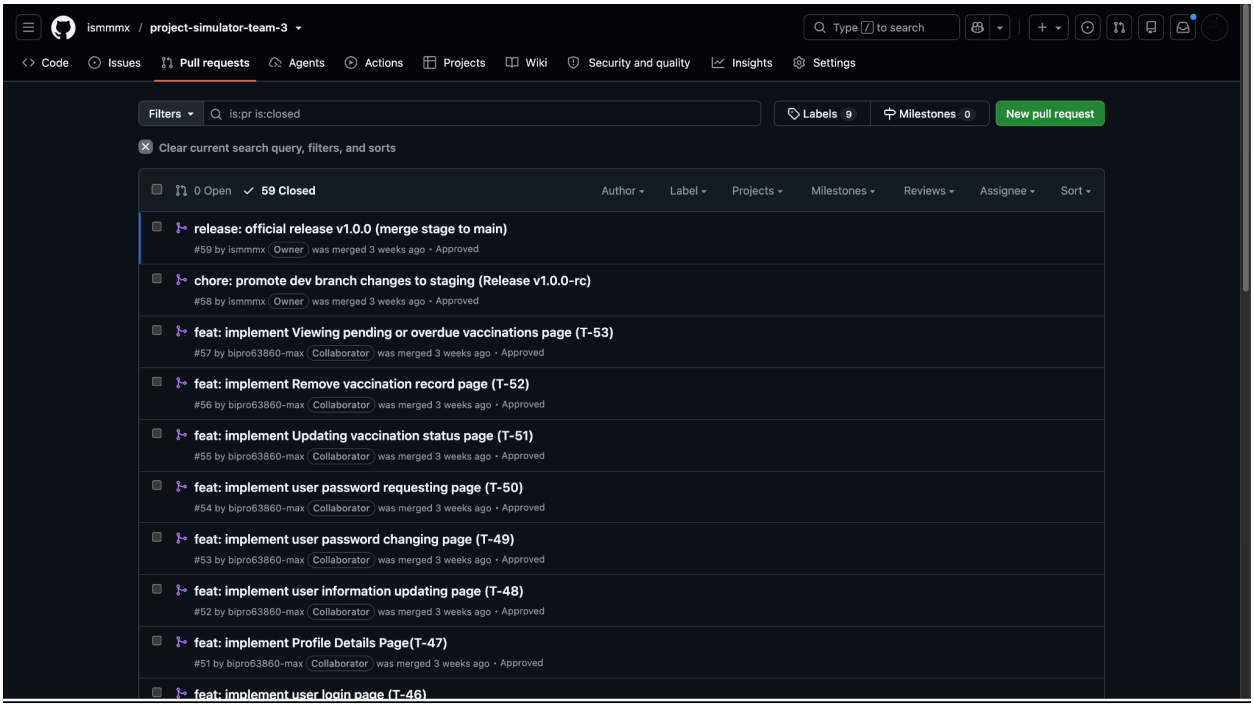


Figure 5.2: Merged Pull Requests Showing Team Code Contributions

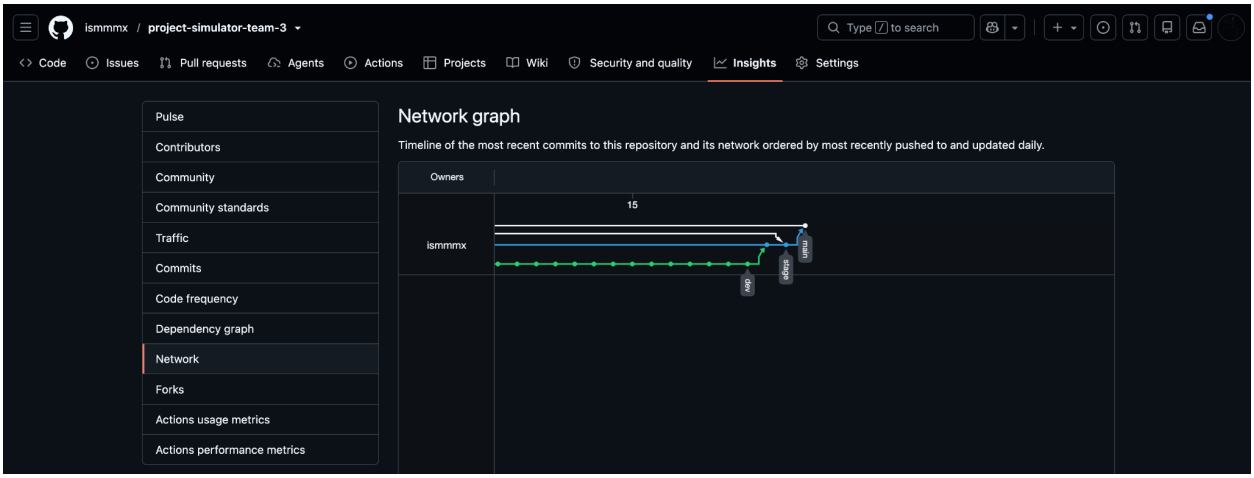


Figure 5.3: Git Network Graph Illustrating Branching and Merge Operations

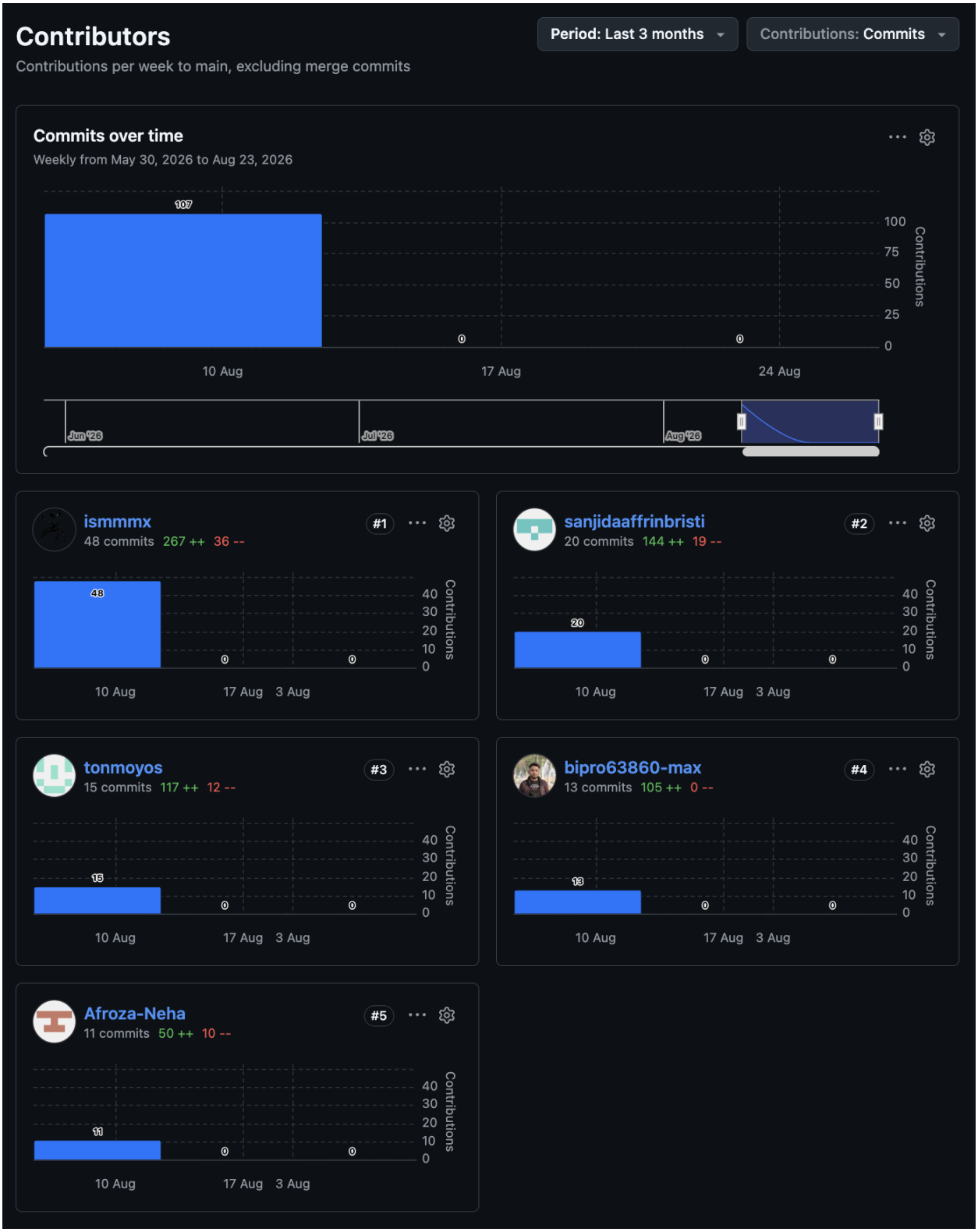


Figure 5.4: Team Contributors Activity and Commit Distribution Graph

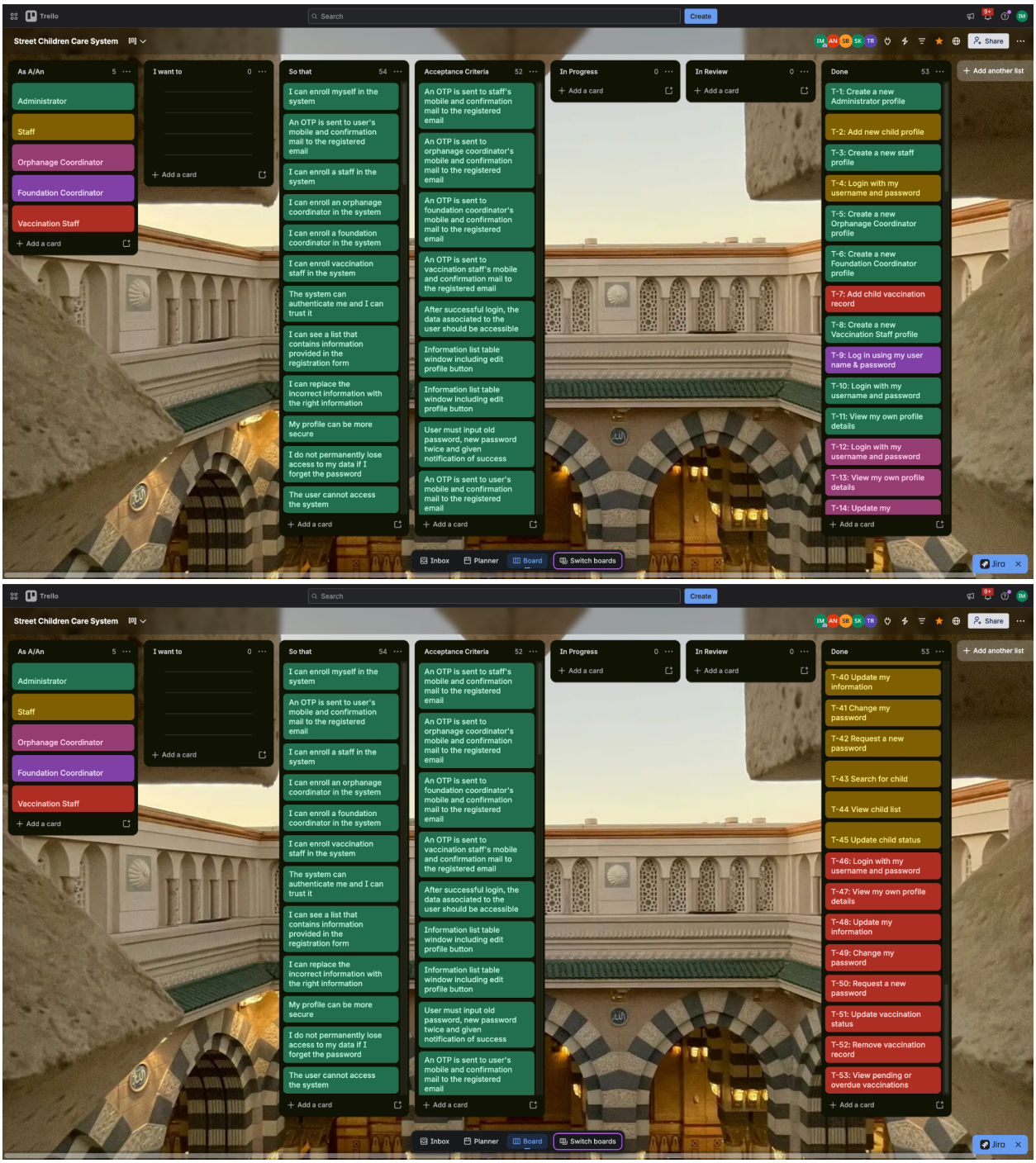


Figure 5.5: Project Task Tracking and Workflow Management on Trello

7. CONCLUSION

The Street Children Care System provides a single platform desktop based solution to reduce the need for manual paper records, avoid duplicate child records and reliably manage child health data. The application is designed in C# Windows Forms with SQL Server connection and integrates five different user roles: Administrator, Staff, Orphanage Coordinator, Foundation Coordinator, Vaccination Staff, in a single secure and role governed application. By using the Extreme Programming (XP) model, our team was able to finish tasks on a weekly basis, use Trello to organize the tasks, and create code quality by utilizing the Git Pull Requests and peer reviews. System testing was performed and it was found to run accurately and securely for child registration, institutional care management, tracking of donations, and vaccination schedules. Overall, the system provides a reliable digital platform to assist street children welfare and enhances the system's administration efficiency.

Source Code Repository (GitHub):
https://github.com/ismmmx/agile-xp-street-children-care-system

UI/UX Design Specifications (Figma):
Street_Children_Care_System – Figma

Project Management Board (Trello):
https://trello.com/b/yeTZYjen/street-children-care-system